

Cross-border offshore hydrogen trade and carbon mitigation in Europe

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Abstract

European countries are pursuing the net-zero transition and offshore energy-resource development. Ireland and the UK have announced 2050 offshore wind ambitions of 37 and 125 GW, more than twice projected domestic power demand, while continental demand centres such as Germany are calling for cleaner fuel resources. Exporting surplus offshore green hydrogen could bridge this supply-demand imbalance and mitigate emissions across Europe. Yet the roles of these island countries are often underestimated when offshore hydrogen is assessed mainly by production cost. Here we develop an integrated European assessment linking cost-supply curves, hourly power-gas operation, hydrogen and ammonia shipping, national demand and external import competition from 2030 to 2050. Results indicate that the UK will be the largest hydrogen supplier from 2030 to 2050, while Ireland nearly matches it in 2050, with both reaching about 162 TWh yr⁻¹ of net exports. This export potential could reduce 175.5 Mt CO₂ yr⁻¹ of carbon dioxide emissions in Europe in 2050. This general flow of hydrogen from the west to the east not only facilitates Europe's net-zero progress, but also reshapes the energy supply structure and helps to ensure energy security across the European continent.

Keywords: hydrogen, offshore wind, carbon mitigation, international trade

1 Introduction

Offshore wind offers a promising solution for achieving net-zero ambitions in Europe, especially for island countries. For example, Ireland and the UK have announced offshore wind development plans reaching 37 and 125 GW of installed capacity, respectively, by 2050 [1, 2]. These targets are highly ambitious because they are more than twice the projected values of their overall power demands in 2050 [3, 4]. At this scale, local power systems cannot fully utilise the resulting offshore wind volumes, leaving surplus generation that increases connection bottlenecks, curtailment risk and real-time balancing pressures.

These countries' significant offshore wind ambitions therefore create an important role for green hydrogen. Produced from renewable electricity through electrolysis, green hydrogen avoids the direct operational emissions associated with fossil-based hydrogen routes [5]. As a storable and transportable energy carrier, hydrogen can absorb offshore wind that cannot be fully accommodated by domestic power systems, reduce curtailment pressure, and move renewable energy across time, sectors and borders. It can be used in domestic industry, heat, power and gas systems, or transported through shipping and gas networks to distant demand centres. This export pathway is important because surplus offshore wind in Ireland and the UK coincides with growing hydrogen demand in continental Europe. Germany expects annual imported hydrogen needs of 45–77 TWh by 2030 [6, 7], and Belgium has identified 20 TWh of imported hydrogen demand by 2030 [8]. Linking exporting and importing countries can ease offshore wind absorption pressures in Ireland and the UK while supplying low-carbon fuels to hard-to-decarbonise sectors across continental Europe.

Cost remains a central factor shaping the large-scale production and utilisation of offshore green hydrogen, and it will strongly influence the development of future international hydrogen trade. The levelised cost of hydrogen (LCOH) in 2030 is projected to be around 54–184 € MWh⁻¹, a range that overlaps with blue hydrogen with carbon capture and storage (CCS) and green hydrogen from onshore wind [9]. In Ireland, offshore wind capacity is expected to exceed onshore wind capacity around 2035 because of richer wind resources, higher capacity factors and fewer land-use restrictions [10]. Previous studies have evaluated offshore hydrogen costs in individual European countries and regions, including North Sea countries [11], the UK [12], Poland [13], and the Iberian Peninsula [14, 15]. These studies provide LCOH projections to 2050, with detailed cost breakdowns, production and transmission routes, and uncertainty analyses. However, few studies compare the cost competitiveness of different countries within a unified framework, which is needed to understand how offshore hydrogen costs may shape the future European hydrogen trading landscape.

European policy documents have identified several major hydrogen import routes, including supplies from North Sea countries (Denmark, Germany and the Netherlands), North Africa, the Middle East and the Americas. By contrast, the export potential of Ireland and the UK, despite their Atlantic offshore wind resources, remains less quantified. Recent Irish and UK policy and consulting reports have recognised green-hydrogen export opportunities [1, 16], but they provide limited quantitative evidence on offshore hydrogen costs, export potential and transport choices. In particular, Irish export assessments have often focused on a pipeline connection to France,

leaving shipping-based hydrogen and ammonia trade underexplored. More broadly, single-country studies cannot determine how European hydrogen imports and exports should be allocated across countries. An integrated analysis is therefore needed to assess how offshore hydrogen trade can allocate renewable resources, contribute to Europe-wide decarbonisation, and inform investment and infrastructure planning.

In this work, we for the first time investigate the roles of offshore hydrogen, especially the export from Ireland and the UK, in the decarbonisation of the entire Europe. First, we developed a unified bottom-up framework to assess the potential and cost-competitiveness of offshore hydrogen produced by major coastal countries in Europe. High spatial-temporal resolution geo-based meteorological data and social-economic data across Europe are used to simulate the wake effect and quantity-cost supply curves of offshore hydrogen for each country. Secondly, using Ireland as an example, we proposed a new coordinated optimisation framework for power and gas systems to evaluate the domestic integration and export potentials of offshore electricity and hydrogen. The power and gas systems are coupled by gas-fired power plants and electrolyzers, where bidirectional energy exchange is allowed to further activate the flexibility of whole energy systems. Various hydrogen integration schemes (including hydrogen blending and pure hydrogen), corresponding physical constraints, and the evolution of power system operating codes (such as inertia, must-run units, and System Non-Synchronous Penetration (SNSP)) are modelled in future scenarios. Based on the cost-supply curves and exporting potential derived above, we then formulate an optimal international trading model to project the hydrogen flows among countries in 2030, 2040, and 2050. We further use a lowest-LCOH allocation as a counterfactual benchmark and vary external import prices to ensure the robustness of the resulting trade pattern. Then, the overall landscape of offshore hydrogen in Europe’s net-zero transition, as well as the contributions from each country, becomes clear. These key findings can further strengthen confidence in the continuous investment in offshore facilities and help the governments with more bespoke offshore policy design.

2 Results

2.1 Cost-competitiveness of green hydrogen production in coastal European countries

We first consider eleven coastal European countries within the assessment scope: Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden, and the United Kingdom (UK). The selection captures countries with large exclusive economic zones (EEZs) for offshore infrastructure development and the principal coastal nodes connected through relatively direct European sea routes. Historical meteorological data for offshore wind-resource modelling, including wind speed, air pressure, and temperature, are obtained from the ERA-5 product at 1 h and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions [17]. Hourly capacity factors and wake-effect losses are simulated using the power curve and deployment assumptions of a 15 MW offshore wind turbine [18]; the turbine power curve and wake-effect setup are shown in Supplementary Figs. 2–4. The resulting wind-speed probability distributions across national EEZs are presented in Fig. 1b. Ireland has the richest wind resource, with

the highest modal wind speed, followed by the UK, largely because both countries are exposed to Atlantic wind conditions. Compared with continental countries, these two island countries show higher probabilities of strong wind speeds and a more even distribution of wind resources across the year.

Using these capacity factors and the assumed development, capital, operating and decommissioning costs of the offshore hydrogen production platform (including wind turbines, foundations and electrolyzers; see Supplementary Figs. 5–10 and Supplementary Tables 3 and 4), we estimate the spatial LCOH distribution shown in Fig. 1a. The model also accounts for bathymetry, port locations, and connection points to the power and gas networks, allowing water depth, maintenance distance and network-connection distance to influence site-level costs [19, 20]. The estimated LCOH ranges from 84.93 to more than 200 € MWh⁻¹ across the studied EEZs. Denmark has the lowest LCOH, supported by high capacity factors and a shallow continental shelf. This lower bound approaches the blue-hydrogen benchmark, although it remains 33.94 % higher than the onshore blue-hydrogen value of 68.92 € MWh⁻¹ [21]. The UK’s LCOH ranges from 85.97 to 307.26 € MWh⁻¹, giving it the second-lowest lower-bound LCOH. Ireland’s LCOH ranges from 90.42 to 201.8 € MWh⁻¹, making it less cost-competitive than the leading countries despite its strong wind resource.

After accounting for wake losses, vessel density, excluded areas (such as military and aquaculture areas), and future technology cost reductions [22], we derive hydrogen cost-supply curves for 2030, 2040 and 2050 (Fig. 2). These constraints affect the effective power density available for offshore wind deployment; the supporting vessel-density, power-density and cost-reduction inputs are shown in Supplementary Figs. 11 and 12. In 2030, the marginal LCOHs of Ireland and the UK are 76.10 and 79.02 € MWh⁻¹, respectively (Fig. 3), with Ireland slightly lower than the UK. At equal production volumes, however, Ireland’s LCOH is generally higher than the UK’s. At low production volumes, Ireland is more cost-competitive than Portugal, Spain, Norway and Sweden, but it becomes less competitive than Sweden once capacity exceeds 10 GW. Within the first 5 GW, the UK remains close to Denmark’s cost level, supporting an early cost advantage for offshore hydrogen production. Beyond this range, the UK still retains lower costs than most other countries. However, because the UK’s offshore wind target is more ambitious, its marginal LCOH ranks eighth. This means that countries with lower marginal LCOHs, such as France and the Netherlands, can still supply limited volumes of competitive hydrogen despite having higher overall cost-supply curves. At this stage, offshore green hydrogen remains above the blue-hydrogen benchmark in most countries.

By 2040, technology and supply-chain learning reduce LCOH by around 21% [22]. As offshore wind capacity targets expand, country cost rankings also shift. Ireland’s marginal LCOH reaches 67.40 € MWh⁻¹ and becomes higher than those of the UK, Sweden and France, because its larger offshore capacity target pushes production into higher-cost sites. The UK moves ahead of Sweden in the ranking because of its favourable offshore conditions, including capacity factors, water depth and coast-line length. Denmark remains the leading cost-competitive country. By 2040, more countries begin to approach the blue-hydrogen cost range, especially Denmark, the Netherlands and the UK. By 2050, the marginal LCOHs of Ireland and the UK fall

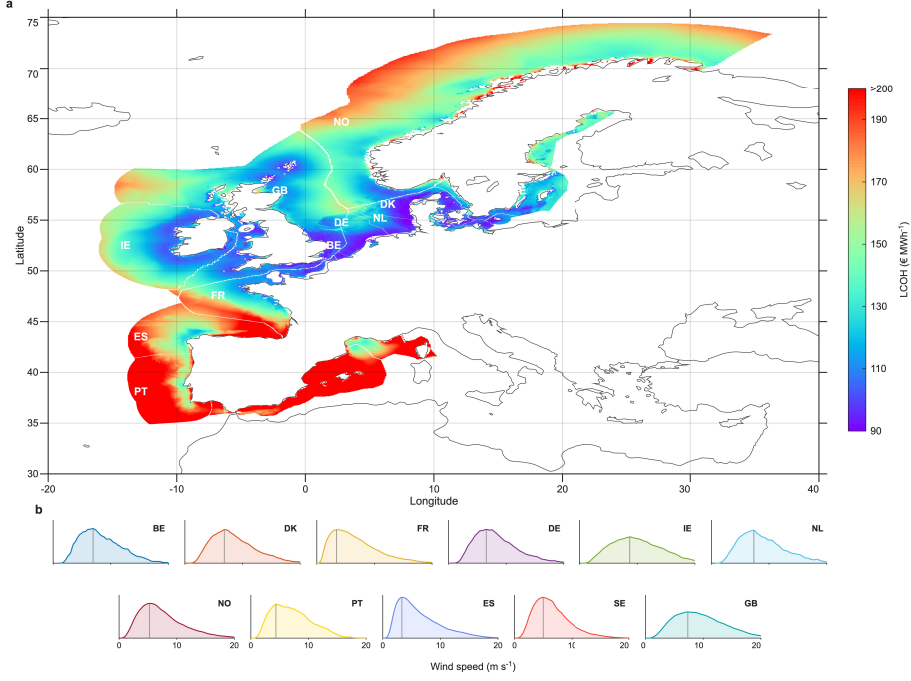


Fig. 1 LCOH map and wind speed in coastal European countries. **a** LCOH map in EEZs of coastal European countries. The black line outlines the EEZs. **b** Probability density of wind speed for each country. The vertical grey line in each figure shows the wind speed most likely to occur. BE, DK, FR, DE, IE, NL, NO, PT, ES, SE, and GB are the abbreviations for Belgium, Denmark, France, Germany, Ireland, The Netherlands, Norway, Portugal, Spain, Sweden, and the United Kingdom, respectively.

further to 54.42 and 52.51 € MWh⁻¹, respectively. The UK keeps its ranking position, whereas Ireland becomes less competitive in terms of both average and marginal LCOHs because of its ambitious offshore capacity target and less favourable offshore conditions. With further LCOH reductions and a relatively stable blue-hydrogen benchmark, all studied countries begin to approach or enter the blue-hydrogen cost range.

Overall, Denmark remains the most cost-competitive country, with the lowest LCOH across the 0–150 GW capacity range. Ireland and the UK occupy intermediate positions in the cost ranking and retain some cost-competitive potential under favourable supply-demand conditions, as assessed in the following sections. By 2050, with a relatively steady blue-hydrogen benchmark [21], offshore green hydrogen approaches or enters the blue-hydrogen cost range across all studied countries.

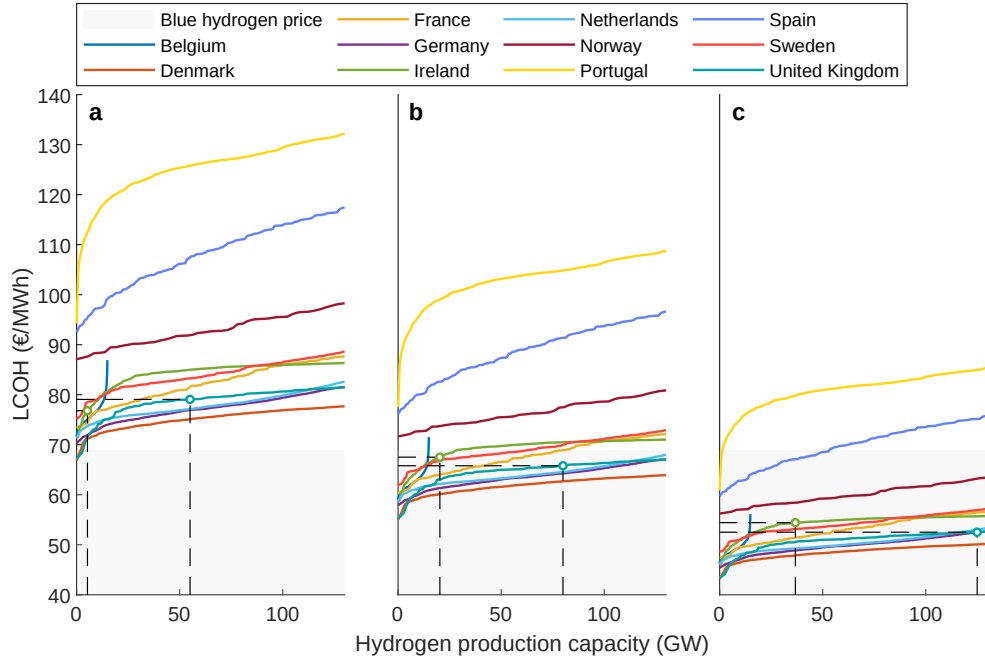


Fig. 2 Offshore green hydrogen cost-supply curves by country and year. a–c, National cost-supply curves for 2030, 2040 and 2050, respectively. Solid lines denote countries. Markers on Ireland’s and the UK’s curves indicate their offshore wind capacity targets. Grey shading shows the projected blue-hydrogen cost range.

2.2 Domestic offshore wind integration and hydrogen export potential

Because the ambitious offshore wind capacity targets in Ireland and the UK are much higher than their power demands, we analyse the integration potential of domestic energy systems to determine the export potential of green hydrogen. A whole energy system optimisation approach is used to simulate operating conditions hour by hour across a whole year with uncertain offshore wind generation. The power and natural gas systems are deeply interconnected through gas-fired power plants and on/offshore electrolyzers, which are optimised simultaneously to expand flexibility for the accommodation of offshore wind fluctuations. Green hydrogen can be blended into the natural gas network at varying ratios, allowing bulk transport and enabling the analysis to cover future scenarios from zero hydrogen to pure hydrogen networks. Hourly unit commitment is conducted considering all available generations in the All-Ireland power system, including coal, waste, peat, natural gas, oil, onshore wind, solar, hydro, and offshore wind. The analysis also accounts for practical operating codes designed by EirGrid [23] for future operation scenarios, such as ramping, interconnectors, inertial constraints, must-run units, and SNSP, together with actual hourly power demand curves (see Supplementary Information, Section 4, for detailed methodologies, including Supplementary Figs. 14–17 and Supplementary Table 5).

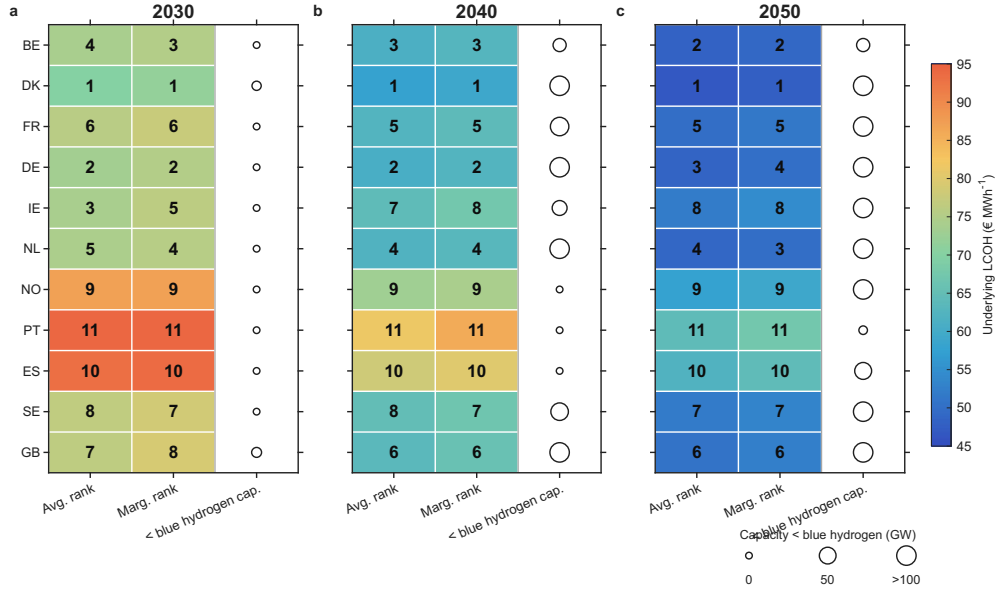


Fig. 3 Country rankings for offshore green hydrogen cost competitiveness. a–c, Cost-competitiveness metrics for 2030, 2040 and 2050, respectively. Cell numbers report average and marginal LCOH ranks among the 11 countries, and colours show the underlying LCOH values. Marginal LCOH is evaluated at each country’s offshore wind capacity target, whereas average LCOH is calculated across the national cost-supply curve. Circles show the hydrogen production capacity with LCOH below the upper bound of the blue-hydrogen benchmark; the smallest and largest circles denote 0 and ≥ 100 GW, respectively.

Figure 4 compares unit commitment results and wind curtailment rates across seasons, model years, and cases with and without export options. Following the UK government’s strategic policy decision to support blending up to 20% hydrogen by volume into Great Britain (GB) gas distribution networks, subject to further enabling decisions, we represent the 2040 case with a maximum 20% gas-network blending ratio [24]. In 2050, pure hydrogen networks are constructed in line with the UK’s hydrogen network roadmap [25]. In summer without export, the average wind curtailment rate generally increases from 67.51% in 2030 to 71.29% in 2050 as offshore wind capacity expands (Fig. 4a–c). This high curtailment is mainly caused by the mismatch between high available wind generation and low demand. In winter, by contrast, the average wind curtailment rate decreases by 56.25% from 2030 to 2050 because lower wind availability makes the higher hydrogen blending ratio the dominant flexibility source, allowing the gas system to absorb more wind.

Even after natural gas is fully replaced by hydrogen and the flexibility of the energy system is fully exploited, a considerable proportion of offshore wind remains curtailed because domestic energy demand is too limited to absorb the available generation. This residual curtailment highlights the need for hydrogen exports. Adding export access substantially reduces offshore wind curtailment across all model years and seasonal scenarios, as shown by comparing the no-export cases (Fig. 4a–f) with the export cases

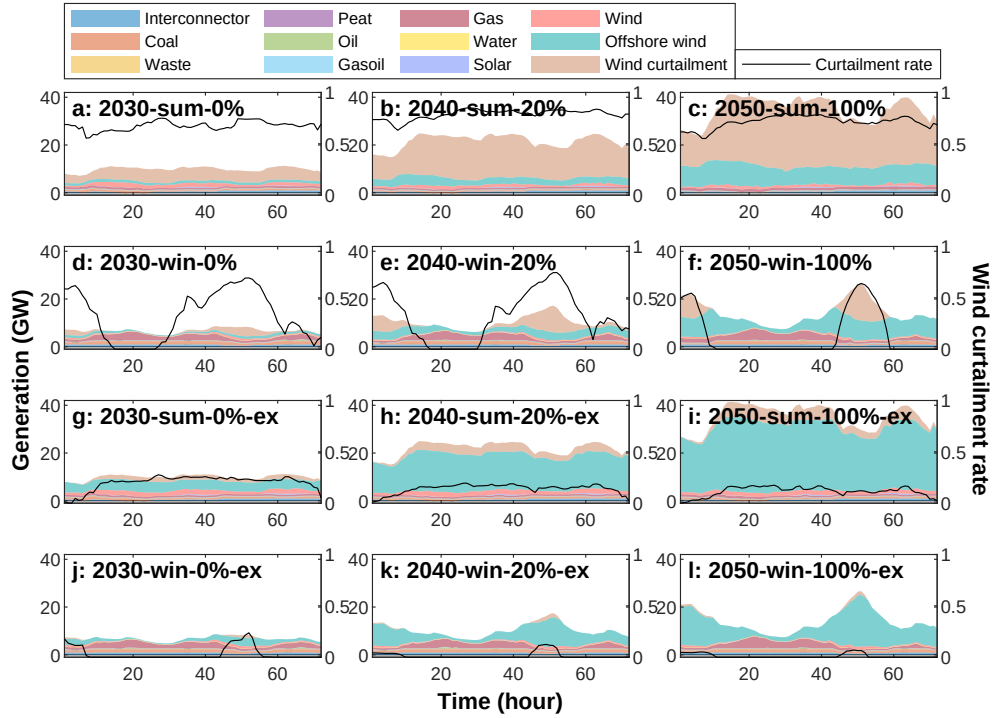


Fig. 4 Power-system operation under different hydrogen-integration and export cases. Panels compare model years, seasons, hydrogen blending ratios and export conditions. The labels 2030, 2040 and 2050 denote model years; *sum* and *win* denote a typical summer day with high wind and low demand and a typical winter day with low wind and high demand, respectively. The labels 0%, 20% and 100% denote the maximum hydrogen blending ratio in the gas system, and *ex* denotes the export case.

(Fig. 4g–l). For example, the average wind curtailment rate in the 2050 summer case falls to 11.53%. Ireland also has substantial export potential, reaching an average of 27 GW in Fig. 4i, equivalent to 1.68 times total domestic energy demand. Offshore-wind uncertainty still requires operating reserves to be scheduled ahead of real-time operation, and reserve costs, reserve-capacity requirements, inertia constraints and must-run unit policies continue to limit the accommodation of marginal offshore wind. However, these operating constraints do not scale proportionally with offshore wind capacity. Once export options are available, the wind curtailment rate therefore decreases as offshore wind capacity increases, which is the opposite of the no-export case.

To extend Ireland’s whole-year result to the wider country set, we estimate domestic offshore-wind absorption in other countries by scaling with electricity and gas demand; the sensitivity of this domestic-absorption assumption is reported in Supplementary Table 7. Figure 5 decomposes annual offshore wind utilisation into domestic power-system use, gas-system use, export availability and residual curtailment. In 2030, the UK has the largest export potential, reaching 178 TWh, because its offshore wind capacity is high relative to domestic load. The Netherlands, Denmark and Ireland follow, with export potentials of 65, 30 and 16 TWh, respectively. In 2040, the

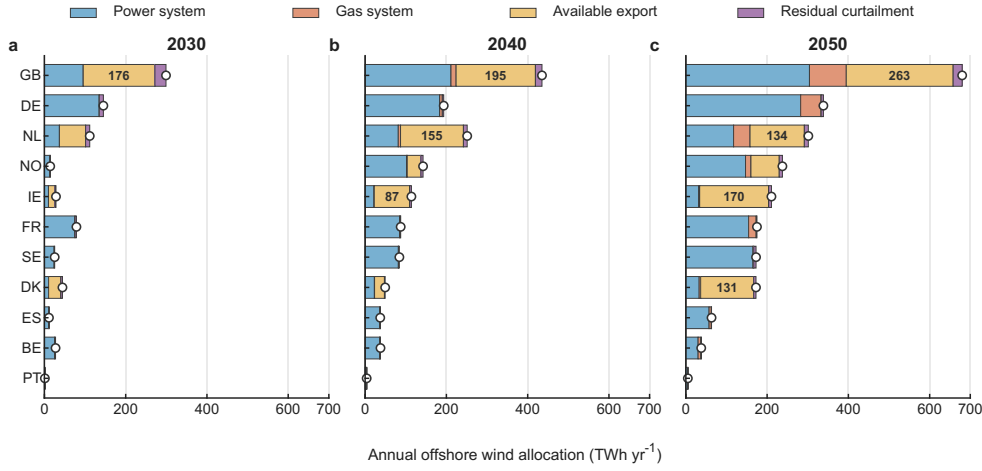


Fig. 5 Annual offshore wind utilisation and export availability. a–c, Offshore wind utilisation in 2030, 2040 and 2050, respectively, decomposed into domestic power-system use, gas-system use, export availability and residual curtailment.

UK and the Netherlands remain the two largest potential exporters, although UK offshore capacity grows more slowly than its energy demand. Ireland’s offshore capacity, however, increases fourfold to 20 GW while its energy demand grows more slowly, raising its export potential to 86 TWh and making it the third-largest potential hydrogen exporter, ahead of Denmark. In 2050, the 100% hydrogen-network case increases domestic offshore-wind absorption, especially in countries that rely more heavily on natural gas generation. As a result, the Netherlands’ export potential decreases by 13.87% even though its offshore capacity increases. Countries with lower offshore potential and higher domestic energy demand, such as Germany and France, have no remaining green hydrogen available for export. By contrast, export potential from the UK and Ireland continues to increase by 32.59% and 96.82%, respectively, relative to 2040. The UK remains the largest potential exporter, with 232 TWh, while Ireland reaches 167 TWh and becomes the second-largest potential hydrogen exporter in Europe because of its large offshore capacity and limited domestic energy demand.

2.3 Carbon emission mitigation potential in Europe

Combining the cost-supply curves and export-potential estimates from the previous sections, we solve an international hydrogen and ammonia trade model to project future cross-border flows. The country-level cost-supply curves are represented by piecewise-linear functions and combined with shipping costs, national demand and export-availability constraints in a cost-minimising optimisation; the optimised deliveries are then used to calculate supplier-attributed carbon-mitigation contributions. The complete model formulation is given in Supplementary Information, Section 5. Demand and shipping-cost inputs are shown in Supplementary Figs. 18 and 19, and the resulting hydrogen-equivalent flows and supplier-attributed carbon-mitigation values are shown in Fig. 6.

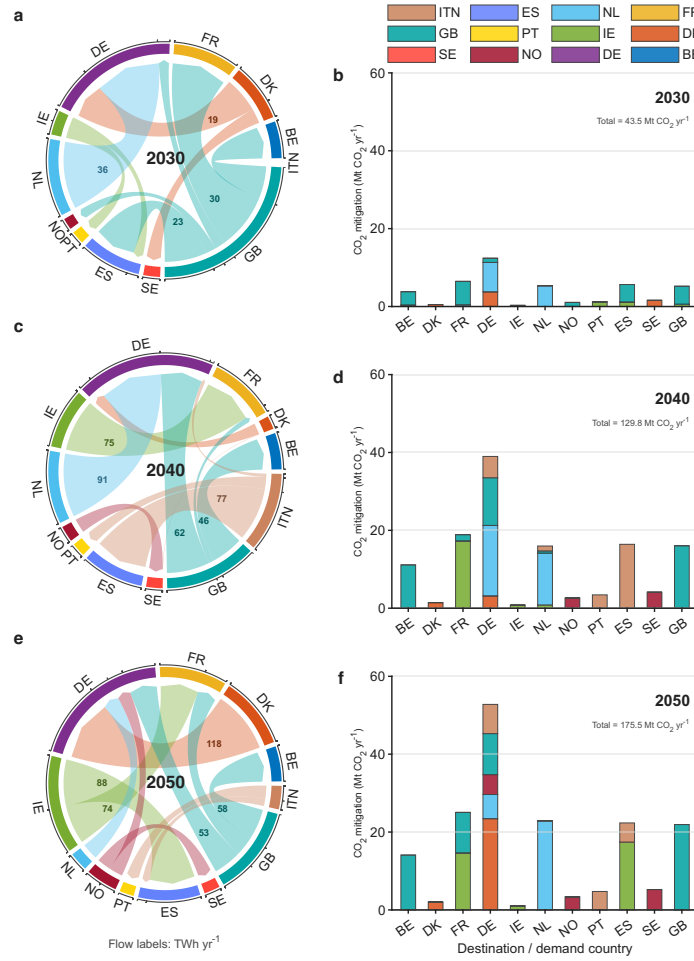


Fig. 6 Sankey diagram of hydrogen flows and contributions to decarbonisation a, c, and e, are the Sankey diagrams of hydrogen flows among these countries and international hydrogen imports in 2030, 2040, and 2050, respectively. ITN represents international hydrogen import. Please note that only the hydrogen flows between countries are drawn. The hydrogen supplied by and to itself is not indicated in these figures. b, d, and f show the contributions to the decarbonisation of each country by international green hydrogen trading. Different bars represent the reduction in carbon dioxide emissions, and different colours indicate different countries' contributions.

In 2030, the UK is the dominant hydrogen exporter, followed by the Netherlands, Denmark and Ireland. UK exports supply 98.29%, 98.38%, 26.08% and 83.99% of the hydrogen demand in Belgium, France, Germany and Spain, respectively. Germany is the main hydrogen receiver because of its high demand. After UK exports, 25.72% and 46.80% of Germany's remaining hydrogen demand is supplied by Denmark and the Netherlands, respectively, owing to their lower production costs, closer locations and consequently lower shipping costs. By 2040, international hydrogen imports become essential because offshore capacity grows more slowly than hydrogen demand. The

UK’s market share declines sharply and it can no longer supply Germany after meeting demand in Belgium and France. As a result, Germany imports 102 TWh of hydrogen from international markets, such as Eastern Europe and South Africa, while Norway, Portugal and Spain also rely partly on international imports. At the same time, Ireland’s role expands, supplying 28 and 52 TWh of hydrogen to France and Spain, respectively. In 2050, as electrification approaches its later stage and hydrogen-demand growth slows, international imports decline but remain important in the overall supply structure, particularly for Germany and Portugal. The supply share previously covered by international imports and the Netherlands is instead taken up by Ireland. Although Ireland’s hydrogen production capacity remains lower than that of the UK, its net exports increase to about 162 TWh yr⁻¹ by 2050, nearly matching the UK and making Ireland one of Europe’s two leading offshore hydrogen exporters.

The trade pattern is also mirrored in the supplier-attributed CO₂-mitigation values shown in Fig. 6b, d and f, with accounting values summarised in Supplementary Table 8. Total mitigation reaches 43.46, 129.77 and 175.50 Mt CO₂ yr⁻¹ in 2030, 2040 and 2050, respectively, equivalent to 0.74, 2.21 and 2.99 times Ireland’s current annual CO₂ emissions [26]. European offshore suppliers account for all of the 2030 mitigation and for 103.19 and 158.30 Mt CO₂ yr⁻¹ in 2040 and 2050, respectively; the remaining mitigation is attributed to the external import node. In 2030, the UK and the Netherlands are the main European offshore contributors to CO₂ mitigation. By 2040 and 2050, similar to the hydrogen supply structure, Ireland becomes increasingly important. Ireland and the UK account for all CO₂ mitigation within their own countries, at 1.05 and 21.93 Mt CO₂ yr⁻¹, respectively, and also support emission reductions in other European countries, contributing 33.45 and 27.21 Mt CO₂ yr⁻¹ in total.

The integrated analysis of offshore hydrogen trade is needed to project realistic future hydrogen trading patterns, beyond what can be inferred from production cost alone. We therefore compare the integrated result with a lowest-LCOH-only counterfactual, in which total European hydrogen and ammonia demand is allocated to offshore supply segments in ascending LCOH order. This cost-only benchmark still uses planned offshore capacity and capacity factors to constrain supply, but it does not account for distance-dependent shipping costs or the market-clearing effects of rising marginal supply costs. The cost-only baseline gives a different exporter pattern from the integrated model (Fig. 7a). It identifies Denmark as the leading net exporter in 2030, the Netherlands in 2040 and Denmark again in 2050, whereas the integrated model gives the UK/Ireland exporter pattern discussed above. In 2050, the cost-only baseline assigns only 41 TWh yr⁻¹ of net exports to Ireland and places the UK in a 44 TWh yr⁻¹ net-import position. This reversal indicates that the strategic-exporter result is not a direct consequence of LCOH ranking, but emerges only when production costs, shipping, demand and market clearing are considered together.

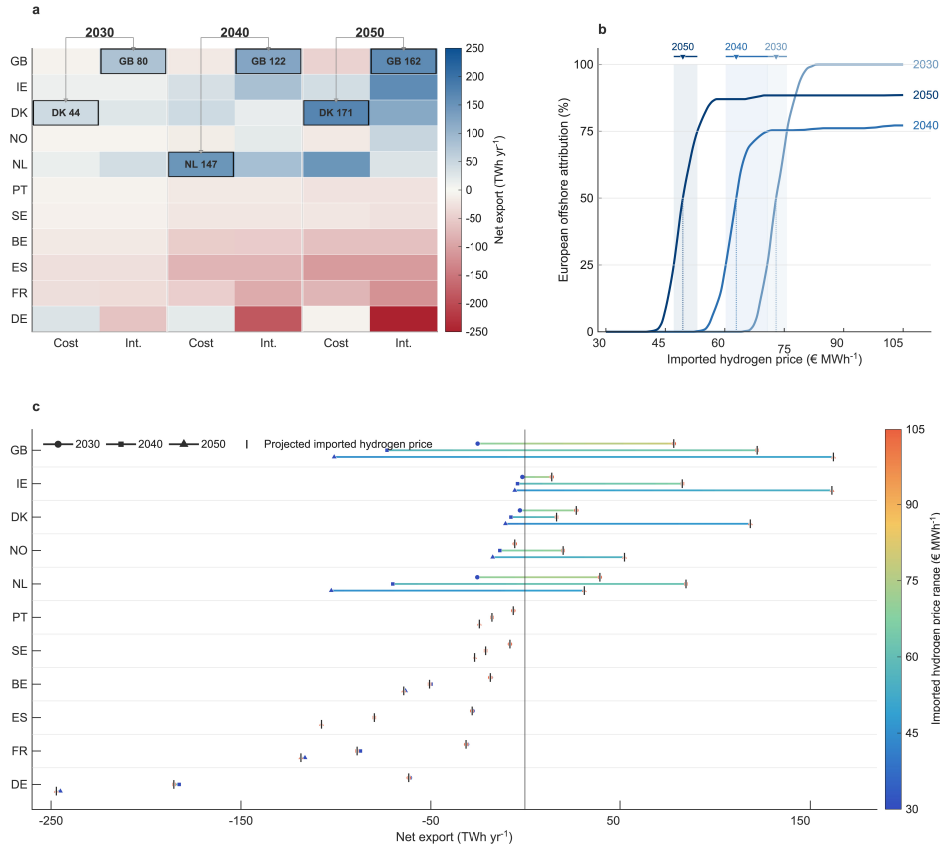


Fig. 7 Counterfactual allocation and import-price sensitivity identify the strategic-exporter pattern. **a** Country net-export balance under a lowest-LCOH-only baseline and the integrated model in 2030, 2040 and 2050. Blue indicates net export and red indicates net import. Black boxes mark the leading net exporter in each model-year combination; labels inside the boxes give the leading exporter and corresponding net export in TWh yr⁻¹. **b** Carbon-mitigation attribution shifts from international imports to European offshore suppliers as the international hydrogen-equivalent import price increases, shown for 2030, 2040 and 2050. **c** Country net-export trajectories across the 30–105 € MWh⁻¹ response range. Each country has three horizontal trajectories for 2030, 2040 and 2050. Line colour indicates the international hydrogen-equivalent import price along each trajectory; endpoint marker shape and line position identify the model year.

The robustness of our analysis is tested through a sensitivity analysis of internationally imported low-carbon hydrogen and ammonia prices. The international hydrogen-equivalent import price is scanned from 30 to 105 € MWh⁻¹. The imported hydrogen price strongly affects the market share of European offshore hydrogen in all

model years (Fig. 7b). Critical thresholds are identified; for example, in 2050, the European offshore hydrogen share increases sharply as the imported hydrogen price rises to 56–60 € MWh⁻¹. The country-level results show the same shift across model years. In Fig. 7c, the line colours show how each country’s net export changes as the import price increases. Net exports from exporting countries, such as the UK and Ireland, are sensitive to the international hydrogen-equivalent import price, while net-importing countries, such as Germany and Portugal, show the opposite trend. The benchmark prices based on the International Renewable Energy Agency (IRENA) projection of green hydrogen in North Africa [27] are marked by short vertical lines in Fig. 7c. These benchmark prices appear near the right end of the country-level lines, indicating that the UK and Ireland are expected to remain in leading exporter positions. Together with Fig. 7b, this result shows that their leading positions are stable as long as the imported hydrogen price is higher than around 50 € MWh⁻¹.

3 Discussion

Our results show substantial potential for offshore green hydrogen in Europe’s net-zero transition, especially the export potential of Ireland and the UK. Denmark has a clear cost advantage in offshore hydrogen supply across the study period, while Ireland and the UK occupy intermediate cost positions despite their strong wind resources. As technology costs decline in the long term, the lowest average LCOH can fall to 46.71 € MWh⁻¹ by 2050. Offshore green hydrogen also approaches or enters the blue-hydrogen cost range across all studied countries, owing to the relatively steady fossil fuel price and CCS infrastructure cost. The domestic offshore wind and hydrogen integration analysis shows that Ireland and the UK not only have cost potential, but also demonstrate considerable exporting potential. Ireland is projected to have the second-largest available hydrogen exporting capacity in Europe by 2050, reaching 167 TWh, because of its ambitious offshore policies and low domestic energy demand. Countries such as Germany have higher offshore capacity, but larger domestic energy demand absorbs their offshore renewable generation and leaves no hydrogen available for export. In return, hydrogen export capacity can also be considered a flexible resource, helping to improve wind utilisation and ease offshore wind connection pressure in Ireland’s domestic power and gas systems. Finally, our international hydrogen and ammonia trade model links the cost-competitiveness and export-potential analyses to predict the future landscape for cross-border green hydrogen trading. Through hydrogen trading, carbon dioxide emissions in Europe can be reduced by 175.5 Mt CO₂ yr⁻¹ in 2050. By 2050, Ireland and the UK will grow into the two largest green hydrogen suppliers in Europe. They account for about 324 TWh yr⁻¹ of net hydrogen exports in 2050, contributing to 51.3% of the modelled carbon dioxide emission reduction for Europe’s net-zero transition.

Although the Irish government has expressed broad support for offshore hydrogen production and export, it remains cautious about Ireland’s competitiveness relative to other European producers. In 2024, Ireland published its future framework for offshore renewable energy policy, together with supporting documents from consulting companies [28]. These documents describe Ireland as broadly competitive globally, but not

as having an outright competitive advantage relative to other European production economies. They also suggest that a considerable volume of surplus economic hydrogen could remain after domestic demand is met, creating an opportunity for exports. Therefore, the Irish government is inclined to explore pipeline transportation options, such as building a pipeline to France from repurposed gas pipelines, to reduce transportation costs and improve economic competitiveness. However, if hydrogen export is limited to pipeline transportation, capacity and flexibility will be constrained. Our results highlight promising alternatives. Ireland has no strong advantage in LCOH and does not have the largest offshore hydrogen production capacity. Yet it can still become one of the largest contributors to European decarbonisation through green hydrogen exports. This outcome reflects the combination of fair cost competitiveness, large export potential and broader hydrogen shortages across Europe. This policy-relevant finding can only be captured through a bottom-up optimisation approach. By combining isolated cost-supply curves with integrated energy-system and international trading models, we identify greater opportunities for Ireland in the future hydrogen market.

These results also depend on currently published offshore goals, with the assumption that the planned infrastructure will be delivered on schedule. Our analysis indicates that international hydrogen imports from other areas of the world, such as Eastern Europe, the Middle East and North Africa, are still needed to meet residual hydrogen demand in 2040 and 2050. These imports may face higher costs or less stable supply conditions. Therefore, despite the ambitious offshore goals already in place, further offshore hydrogen expansion remains valuable. The import-price sensitivity analysis supports this interpretation: once the international hydrogen-equivalent import price exceeds about 50 € MWh⁻¹, the UK/Ireland exporter pattern remains stable. First, more ambitious offshore hydrogen production goals could help cover the 62 TWh yr⁻¹ residual international hydrogen import in 2050, equivalent to roughly 14 GW of offshore electrolyzers and corresponding offshore wind farms. Based on our analysis in Section 2.1, most studied coastal countries except Belgium still have potential to further increase offshore capacities. This is especially recommended for countries with lower marginal costs, such as Denmark and the Netherlands. Second, governments could accelerate efforts to address non-technical bottlenecks in offshore infrastructure construction. These include supporting domestic and international investment in industrial manufacturing capacity for offshore wind turbines, electrolyzers, subsea cables and hydrogen shipping vessels. Governments will also need clear and stable policy frameworks to de-risk private investment, accelerate consenting processes for offshore wind farms and establish dedicated hydrogen export corridors. These measures could reduce temporary reliance on international hydrogen imports in 2040 and support a smoother expansion of the hydrogen market.

These quantitative findings could help European governments revisit offshore goals as they update national hydrogen strategies. Our results reveal both promising hydrogen-market potential and continuing shortages in offshore infrastructure investment, which could help attract further investment. This west-to-east offshore hydrogen export flow will significantly reshape the energy-flow pattern in Europe. It can make wider contributions to the net-zero transition and enhance the energy security of

continental European countries that still rely heavily on international gas imports, ultimately helping to address the energy trilemma.

4 Methods

4.1 Techno-economic modelling of offshore hydrogen cost-supply curves

Offshore wind generation and offshore hydrogen production costs are assessed within the EEZs of the selected coastal European countries, i.e., Belgium, Denmark, France, Germany, Ireland, Netherlands, Norway, Portugal, Spain, Sweden, and the UK [29]. Aquaculture sites, Special Areas of Conservation, and military areas are excluded from the available offshore hydrogen production areas, with geospatial boundaries obtained from the European Marine Observation and Data Network (EMODnet) [20]. Within these geographical boundaries, vectorised wind velocity data and other weather variables affecting wind generation, including temperature and air pressure, are obtained from the ERA-5 product of the fifth-generation European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis. Data are provided at 1 h and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions, equivalent to about 45 km by 56 km in southern Europe, over the selected five-year historical period [17].

Wind power generation and capacity factors are calculated hourly using the power curve of a National Renewable Energy Laboratory (NREL) reference 15 MW wind turbine, with technical parameters described in [18]. The power curve is specified at 1 m s^{-1} wind-speed resolution. For fractional wind speeds, linear interpolation is used to estimate wind-turbine power output. Turbine layouts are optimised by balancing power density against interference from wake effects. The wake effect of a single turbine is simulated using the Jensen-Gaussian model, whose effectiveness has been demonstrated in [30]. When characterising cumulative wake effects from multiple upstream wind turbines, wake effects can overlap. Previous studies usually consider only a single wake layer or use enumeration-based methods, which increase the computational burden exponentially with the number of mutually interfering wind turbines and the size of the historical dataset [31, 32]. We therefore use a recursive-tree-based method, which reduces simulation time by 79% with less than 1% loss of accuracy in practice [33]. The specific methodologies and parameter values are provided in Supplementary Information, Section 2.

The offshore hydrogen production LCOH is calculated using a bottom-up approach with four cost modules: development expenditure (DEVEX), capital expenditure (CAPEX), operational expenditure (OPEX), and decommissioning expenditure (DECEX). These cost modules are further broken down into different terms and components, as shown in Supplementary Figs. 6 and 9 and Supplementary Tables 3 and 4. Bespoke techno-economic parameters for Europe are obtained from Offshore Renewable Energy Catapult, including environmental survey costs, wind-turbine costs and cable costs [34]. The geolocation-based LCOH calculation uses five additional inputs: water depth, capacity factor, distance to the nearest port, distance to the grid or gas-network connection point, and distance between wind turbines. Water-depth data within the EEZs of the studied countries are obtained from the bathymetry database

published by EMODnet [19]. The data are converted and applied to grid cells with $0.001^\circ \times 0.001^\circ$ spatial resolution, equivalent to about 0.11 km by 0.06 km near the UK. Water depth affects the wind-farm structure and the amount of material used for the platform. For water depths less than 30 m and between 30–60 m, monopile and jacket fixed-bottom structures are evaluated. Floating turbines are also evaluated in deeper water environments above 60 m, where installation and maintenance costs are higher. Within the EEZs of the studied countries, North Sea areas are relatively shallow and can be deployed with fixed-bottom wind turbines, whereas water depths around the Iberian Peninsula increase faster with distance from the coast. The minimum distance to the port affects the installation cost when shipping materials to offshore locations, as well as the annual maintenance expenses. The major port and city locations in the studied countries are obtained from EMODnet [20]. The minimum distance to connecting points determines the capital cost of export cables that transmit electricity generation from offshore to onshore substations. Distances between wind turbines are determined based on the principle that the maximum interference from adjacent wind turbines from the wake effect should be less than 5%. It affects the cost of array cables that collect the generations from individual wind turbines and converge on the offshore substation.

In the offshore electrolysis platform, several components consume energy from offshore wind, including water pumps and desalination, electrolyzers, hydrogen compressors, and standby batteries. The techno-economic characteristics of the Alkaline Electrolyser (AEC) are summarised in Supplementary Table 4. The AEC system is currently the most mature and durable among these three options, and is therefore used as the reference electrolyser technology. The data on the capital cost and electrical efficiency were obtained from projections by the Offshore Renewable Energy Catapult and the UK’s National Hydrogen Strategy. The efficiencies of AEC are projected to increase from 0.79 in 2030 to 0.82 in 2050 [35]. The capital cost includes previously mentioned key components, as well as necessary auxiliary systems such as platforms and pipelines. Annual OPEX includes maintenance, stack replacement, and labour costs are expressed as a percentage of initial capital investments. Because offshore wind fluctuates and does not always match the electrolyser operating point, electrical and hydrogen storage facilities are included. Water pumps, desalination systems and hydrogen compressors are sized according to hydrogen production. Their total power consumption therefore depends on the hydrogen production capacity and is matched to the rated wind-generation power with a margin. On this basis, a set of algebraic equations (Supplementary Information, Section 3.3) is formulated by maximising the utilisation rate to determine the optimal size of the electrolyser and auxiliary components. Future operating parameters and costs are taken from [34]. The offshore hydrogen production LCOH is then calculated by accounting for the DEVEX, CAPEX, OPEX and DECEX of these devices.

In addition to the LCOH map, hydrogen cost-supply curves require the available hydrogen supply quantity at each LCOH level. The potential installed offshore wind capacity for each grid cell is calculated by multiplying the turbine’s rated power by the maximum number of turbines that can fit into the cell. Distances between wind turbines are determined as described above to control wake effects below the threshold

while maximising power density. Vessel density affects the power density of a wind-farm layout and is obtained from EMODnet at 1 km resolution [36]. Power density is assumed to decrease linearly with vessel density by up to 50% [37]. Three future scenarios for LCOH reductions (high, median, and low) are considered based on expert elicitation surveys [22]. Compared with 2020, the reduction rates in 2030, 2040, and 2050 in the example of median scenarios are projected at 21%, 35%, and 49% for fixed wind farms, and 2%, 17%, and 40% for floating wind farms, respectively.

4.2 Domestic offshore wind and hydrogen integration model

This module estimates how much offshore wind generation can be absorbed by the domestic energy system and how much hydrogen remains available for export under each model year. Offshore power generation is integrated into the domestic power system via marine cables, while for bulk utilisation, hydrogen production is connected to the domestic natural gas system via pipelines through hydrogen blending, with blending levels assumed to range from 0% to 100% across the 2030, 2040 and 2050 model years. Hydrogen blending refers to injecting hydrogen into existing natural gas pipelines, where it is mixed with natural gas and transported to other locations for further use. Compared with local hydrogen utilisation or dedicated hydrogen pipelines, hydrogen blending provides a cost-efficient interim solution for scaling hydrogen demand. Hydrogen blending has received substantial support and has been demonstrated in numerous countries, including the European hydrogen backbone [38], HyDeploy in the UK [39], and HyBlend in the US [40]. The UK’s Department for Energy Security and Net Zero permitted distribution-level hydrogen blending in December 2023, and is actively seeking more evidence to extend it to transmission level [24]. According to the Energy Network Association, the UK aims to achieve 100% hydrogen networks by 2050 [41]. Therefore, the utilisation and integration of offshore power and hydrogen production are constrained by national energy systems, including demand, network transmission capacity and operating schemes. To evaluate the potential of domestic energy systems to accommodate offshore resources, especially for island countries such as Ireland with abundant offshore resources and limited domestic energy demand, we develop a coordinated optimal operation model (COM) for the national energy system, including both power and gas networks. The optimisation is conducted hourly throughout the year to simulate the lowest-cost operating state while maximising offshore resource integration.

The power system module in the COM constitutes a unit commitment framework to schedule the resources in the day ahead for balancing the variation of energy demands and uncertain renewable generation on the nodal scale. The unit commitment cost includes fuel costs and start-up costs for thermal units. For the All-Ireland power system, the practical operational policy roadmap from 2023–2030, as outlined by Eir-Grid and the System Operator for Northern Ireland (SONI), is incorporated in the model, such as inertia constraints, minimum number of conventional units and System Non-Synchronous Penetration (SNSP) [42]. The flexibility constraints for generators are incorporated, including unit ramping limits, minimum online-offline times, minimum output levels and must-run constraints. The power network congestion, which may also hinder the integration of offshore wind generation, is modelled using the DC

power flow model. The dynamic rating of power lines with temperature variations is considered in summer and winter scenarios. Future construction plans for interconnections are considered, such as the 700 MW Celtic Interconnector with France in 2027 [43].

The natural gas system module minimises gas-system operating cost while accommodating hydrogen injection from offshore hydrogen production. With stochastic hydrogen injection, the gas composition and corresponding physical properties, such as heat value and specific gravity, become uncertain, creating gas-security and operational challenges. We define a gas security region according to the updated Gas Safety (Management) (Amendment) Regulations 2023 [44], using relative density, Wobbe index and Weaver flame-speed factor to limit the hydrogen fraction. The Weymouth equation describes the relationship between gas flow rate and pressure drop along a gas pipeline to avoid over- or under-pressure issues in the network. Because variable gas properties make the Weymouth equation more nonconvex, we use a reformulation and solution algorithm comprising bespoke McCormick envelopes, second-order cone reformulation and sequential convex programming. The resulting optimisation problem can be solved robustly and efficiently by off-the-shelf commercial solvers such as Gurobi. Compared with general nonlinear solvers such as IPOPT, computation efficiency improves by more than 90% [45]. When multiple pipelines intersect, different gas compositions are assumed to mix evenly at the junction. This process is also incorporated in the model and generates bilinear constraints that are addressed using the same solution framework (mathematical models are provided in Supplementary Information, Section 4.2).

For an accurate description of power and gas system operations, high-resolution and long-span datasets are used for power generation, consumption and transmission. On the generation side, detailed operational parameters of All-Ireland generators, including capacity and piecewise heat-rate curves, are obtained from the Single Electricity Market (SEM) Committee, which has been used and validated in EirGrid publications [46]. For non-gas fossil fuel units such as coal and oil, commodity cost and heat value are based on the SEM model. For gas-fired units, fuel cost is counted on the gas-system side, and heat value is modelled dynamically with changing hydrogen fractions. For onshore wind and solar generation, wind speed and solar radiation are obtained from ERA-5, and the Beckman model is used to calculate solar output. Hydro generation data are obtained from historical operations [47]. For the consumption sector, future gas demand is predicted according to the Gas Forecast Statement by Gas Network Ireland [48], with cross-sector considerations including the decommissioning of coal-fired power plants (for example, Moneypoint units), the 80% renewable-generation ambition in the Irish government’s Climate Action Plan, the impacts of banned new gas-boiler connections, GDP effects on residential, industrial and commercial demand, and the construction of new compressed natural gas stations. As a result, gas demand increases in the short term, reaches a peak around 2025, and then continues to decrease. Similarly, daily electricity peak demand up to 2032 is predicted using the bottom-up approach in EirGrid’s Ten-Year Generation Capacity Statement [49]. This dataset covers forecast values through 2032 in typical seasonal scenarios (winter, summer, and spring-autumn). Because annual power demand increases

steadily, the Autoregressive Integrated Moving Average (ARIMA) method is used to fill the missing values between 2033 and 2050. Fifteen-minute-resolution electricity and gas demand curves are used for daily operational analysis and are obtained from EirGrid’s smart grid dashboard and the Gas Network Ireland data dashboard in different seasonal scenarios [50, 51]. For the power transmission system, high voltage power lines over 110 kV are modelled, and the network topology, geographical locations, and electrical parameters (such as impedance) are obtained from EirGrid’s Transmission Forecast Statement [52]. The high-pressure gas transmission pipeline network (above 16 bar) is modelled, and similarly, the network topology, geographical locations, and pipeline length are obtained from Gas Network Ireland’s Network Development Plan [53]. Friction factors and diameters are assumed according to a dedicated case study of Ireland [54]. The capacities of power lines are assumed to grow annually at the same rate as the demand growth.

4.3 International hydrogen and ammonia trade model

Besides domestic consumption of hydrogen and ammonia products, surplus supply can be transported as liquid products by shipping to mainland Europe. We therefore propose an international hydrogen and ammonia trade model that uses the supply-cost curves derived above to simulate hydrogen-equivalent trade flows among the studied European countries in 2030, 2040 and 2050. Supply potential and demand define the main boundaries of the future trade landscape. On the supply side, countries with explicit offshore wind targets for all or selected model years, such as Ireland [55] and the UK [56], use those values as maximum offshore wind capacities. For countries without complete offshore wind targets across all model years, such as the Netherlands and Spain, tentative goals from the European Union’s non-binding agreement on offshore wind are used [57]. The hydrogen and ammonia demand is predicted by the European Hydrogen Backbone Project [58].

The objective of the proposed international hydrogen and ammonia trade model is to minimise production and transportation costs while meeting hydrogen and ammonia demand, as detailed in Supplementary Information, Section 5. Production cost is represented by the special-ordered set interpretation of supply-cost curves. Transportation cost includes annualised ship investment costs and operating fuel costs. The annualised ship cost determines the number of ships invested in each possible route, which further determines the maximum trade capacity among countries. The fuel cost is related to the length of the shipping route between the ports of exporting and importing countries from [20]. Due to the discontinuity of the hydrogen supply curves, the original problem is formulated as a mixed integer linear programming problem, and is solved by the Gurobi commercial solver via MATLAB.

To compare the integrated trade result with a production-cost-only allocation, we construct a lowest-LCOH counterfactual in which total European hydrogen and ammonia demand is allocated to offshore supply segments in ascending LCOH order. This baseline is constrained by planned offshore wind capacity and mean capacity factor, but excludes domestic absorption, shipping distance, route capacity and market clearing. Country-level net export is calculated as allocated offshore supply minus domestic hydrogen and ammonia demand.

For the import-price sensitivity analysis, the external import node is assigned a hydrogen-equivalent price from 30 to 105 € MWh⁻¹ in 1 € MWh⁻¹ increments, and the trade model is resolved for each price and model year. The reported outputs are extracted from the optimised production, trading and carbon-attribution variables. Benchmark imported-hydrogen price markers are derived from the IRENA delivered-import to domestic-production cost ratio for North African green hydrogen supplied to Germany [27], applied to the average Ireland and UK marginal LCOH in each model year.

Acknowledgments

This work was supported by the Newcastle University Academic Track (NUAcT) Fellowship.

Author contributions

S.W. conceived the study, developed the modelling framework and MATLAB code, performed the analysis, prepared the figures and wrote the original manuscript. M.M.B. contributed to the energy-system interpretation, policy context and manuscript review and editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The public data sources used in this study are cited in the References and described in the Methods and Supplementary Information. Derived source-data tables generated by the analysis, including the headline-number audit table used to trace manuscript values and a source-data coverage inventory, are available in the project repository under `results/tables/`. Final figure files are provided under `manuscript/figs/`. Large public raw climate, bathymetry and geospatial inputs and large intermediate MATLAB checkpoint files are not redistributed through the GitHub repository because of file-size constraints; the raw inputs are available from the cited public data providers and the checkpoints can be regenerated from the provided code.

Code availability

MATLAB code used in this study is available on GitHub at <https://github.com/ShengWang-EE/HyExport>.

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